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```
import pandas as pd

# Create DataFrame
property_data = pd.DataFrame({
    'Property_ID': [101, 102, 103, 104, 105],
    'Location': ['Chennai', 'Bangalore', 'Chennai', 'Hyderabad', 'Bangalore'],
    'Bedrooms': [3, 5, 2, 6, 4],
    'Area_sqft': [1200, 2000, 900, 2500, 1800],
    'Listing_Price': [5000000, 8500000, 4000000, 9500000, 7000000]
})

print("Property Data")
print(property_data)

# 1. Average listing price in each location
print("\nAverage Listing Price by Location")
print(property_data.groupby('Location')['Listing_Price'].mean())

# 2. Number of properties with more than four bedrooms
count = property_data[property_data['Bedrooms'] > 4].shape[0]
print("\nProperties with more than 4 Bedrooms:", count)

# 3. Property with the largest area
largest = property_data.loc[property_data['Area_sqft'].idxmax()]
print("\nProperty with Largest Area")
print(largest)
```

```
Property Data
  Property_ID  Location  Bedrooms  Area_sqft  Listing_Price
0          101    Chennai         3      1200      5000000
1          102  Bangalore         5      2000      8500000
2          103    Chennai         2       900      4000000
3          104  Hyderabad         6      2500      9500000
4          105  Bangalore         4      1800      7000000
```

```
Average Listing Price by Location
Location
Bangalore    7750000.0
Chennai      4500000.0
Hyderabad    9500000.0
Name: Listing_Price, dtype: float64
```

```
Properties with more than 4 Bedrooms: 2
```

```
Property with Largest Area
Property_ID      104
Location         Hyderabad
Bedrooms         6
Area_sqft        2500
Listing_Price     9500000
Name: 3, dtype: object
```

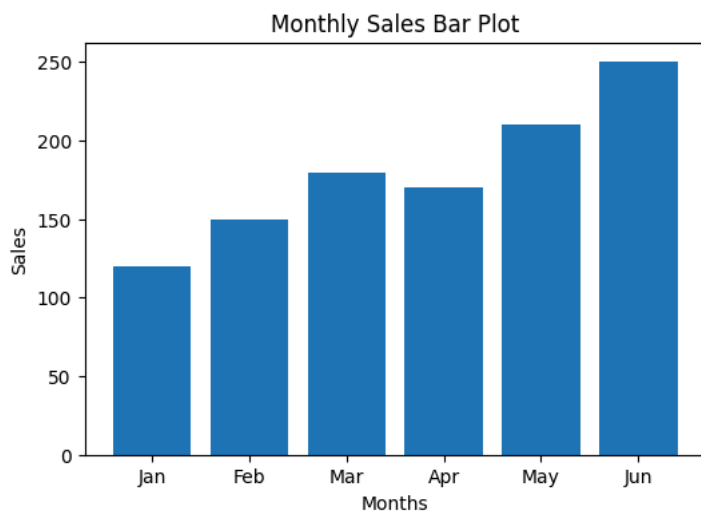
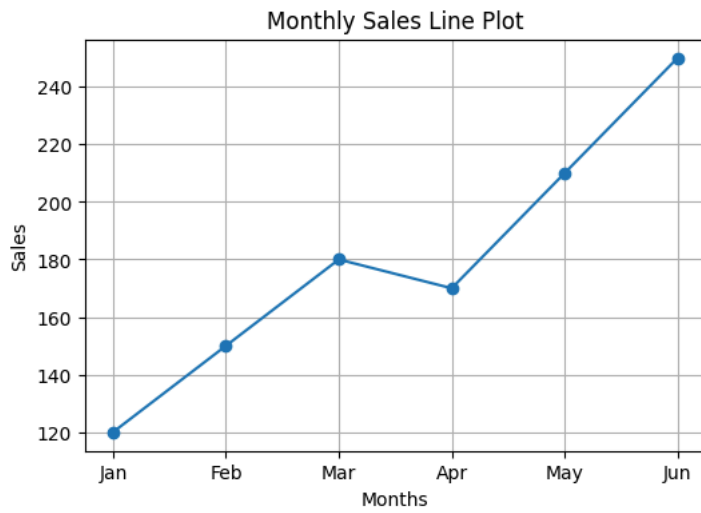
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```
import matplotlib.pyplot as plt

# Monthly sales data
months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [120, 150, 180, 170, 210, 250]

# Line Plot
plt.figure(figsize=(6,4))
plt.plot(months, sales, marker='o')
plt.title("Monthly Sales Line Plot")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.grid(True)
plt.show()

# Bar Plot
plt.figure(figsize=(6,4))
plt.bar(months, sales)
plt.title("Monthly Sales Bar Plot")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.show()
```

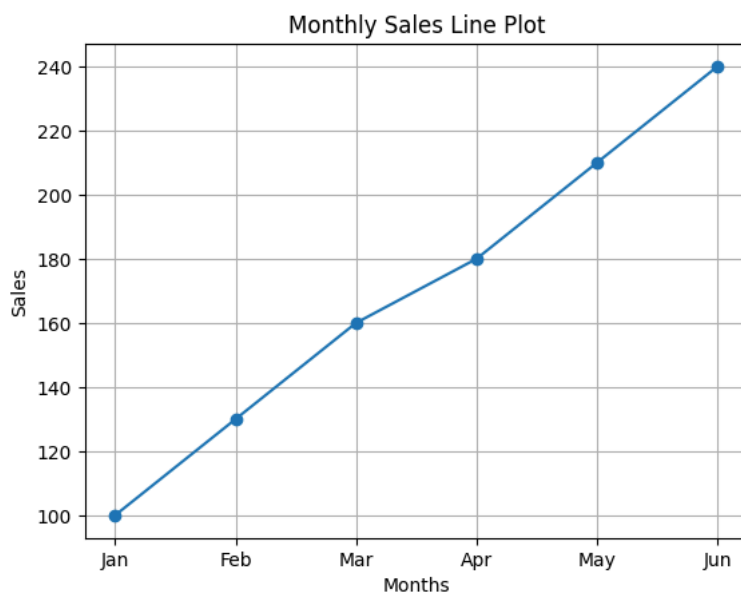


11.1

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [100, 130, 160, 180, 210, 240]

plt.plot(months, sales, marker='o')
plt.title("Monthly Sales Line Plot")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.grid(True)
plt.show()
```

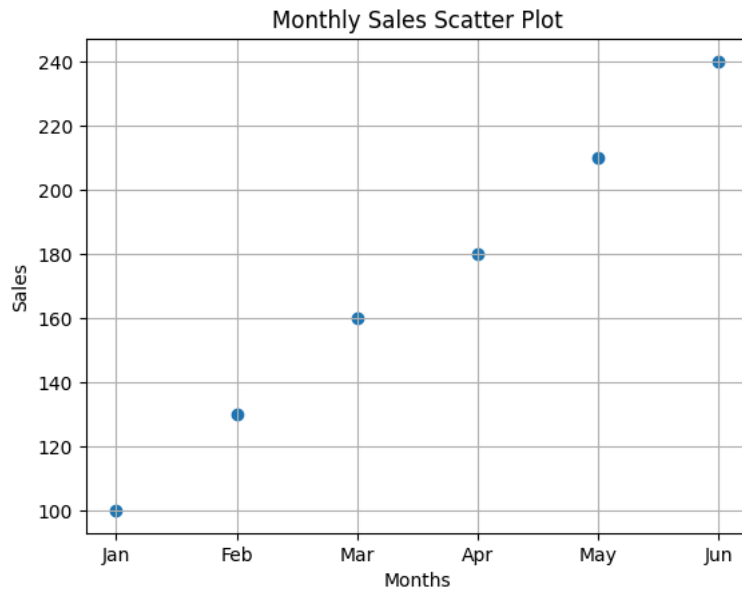


11.2

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [100, 130, 160, 180, 210, 240]

plt.scatter(months, sales)
plt.title("Monthly Sales Scatter Plot")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.grid(True)
plt.show()
```



11.3

```
import matplotlib.pyplot as plt

months = ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun']
sales = [100, 130, 160, 180, 210, 240]

plt.bar(months, sales)
plt.title("Monthly Sales Bar Plot")
plt.xlabel("Months")
plt.ylabel("Sales")
plt.show()
```

